

Maksim Ryzhov

Full-stack Product Engineer · AI & Automation

Open to remote engineering roles · UTC+5 | [GitHub](#) | [LinkedIn](#) | [Telegram](#) | maksryzhov16@gmail.com

Live products: [tunika-tkani.ru](#) · [art-square.studio](#) · [kalpak.dev](#)

PROFILE

Product engineer shipping multilingual production sites, developer tools, and LLM agent systems. I work from observable behavior backward: inspect the running system, make a bounded change, add focused tests, and verify the result in the browser or delivery environment. Page 1 is product work; page 2 is the AI and agent side.

SELECTED PRODUCT WORK

ART SQUARE STUDIO [Multilingual studio site](#) · [private source](#) [art-square.studio](#)

Role: Full product engineering: architecture, UI, content system, security headers, release pipeline.

- Four production locales (RU/EN/KY/DE) with an automated gate that fails the release on any missing translation key.
- 13-check release pipeline — strict nonce-based CSP, accessibility, Lighthouse, real Safari — writing inspectable evidence on every run.

kalpak.dev [Storefront bureau for Telegram shops](#) · [private source](#) [kalpak.dev](#)

Role: Full product engineering and design: concept, UI, motion, release verification.

- Dark editorial storefront on Next.js RSC and Cloudflare Workers with eight switchable live design directions.
- Honesty built into the interface: the one real client project and the demo concepts are explicitly labeled apart.

Tunika [Live commerce platform](#) · [private source](#) [tunika-tkani.ru](#)

Role: Product engineering across responsive UI, catalog publication, operational safeguards, release verification.

- Delivered catalog and product-page behavior across desktop and short mobile viewports, including filters, sorting, badges, fabric units, and JPG/WebP/AVIF selection.
- Hardened Go-backed publication and reconciliation so incomplete output does not replace the known-good catalog; acceptance checks the public product after release.

Mansara [Interactive residential experience](#) · [private source](#)

Role: Frontend interaction architecture, responsive experience, asset validation, browser QA.

- Connected a 120-image building turntable, nine facade views, floor selection, a sample apartment journey, and a six-stop park walkthrough into one stateful browser flow.
- Kept the presentation evidence-bounded with explicit demo data, responsive fallbacks, and no claim of continuous real-time 3D.

PUBLIC DEVELOPER TOOLS

Viewport Sentinel — TypeScript CLI and GitHub Action for deterministic page-by-viewport Chromium checks with JSON and SARIF evidence. Published on npm as [viewport-sentinel](#).

Hermes Agent for VS Code — review-first VS Code client for a self-hosted agent bridge, with file, selection, screenshot, and code-edit context.

ENGINEERING PRACTICE	TECHNOLOGY
Responsive UI · deterministic tests · browser-visible QA · release gates · rollback-aware delivery · explicit product limitations	TypeScript · React · Next.js · JavaScript · Node.js · Go · Playwright · GitHub Actions · HTML/CSS

Daily toolchain: Claude Code · OpenAI Codex · MCP servers and tools · self-hosted agents

HOW I WORK WITH LLMS

I build the plumbing around models rather than the models themselves: agent backends, MCP tools, prompt and context pipelines, and the gates that decide whether an agent's output is allowed to land. Same principle as the product work — every step leaves inspectable evidence, and nothing ships on a model's word alone.

AGENT SYSTEMS

Hermes Agent Self-hosted LLM agent · VS Code client public

[hermes-vscode](#)

Role: Agent backend, transport, review flow, VS Code client.

- Telegram as the control surface: files, selected code, and screenshots go in as context; edits come back as a diff to review before anything touches the working tree.
- Platform side in Python and aiohttp — TTL bearer tokens, rate limiting, SQLite audit trail, systemd units — so a self-hosted agent stays accountable for what it did.

ADW Agentic development runner · private source

Role: Runner architecture, isolation model, quality gates.

- Splits agent work into plan, build, and fix stages, each task in its own Git worktree, so parallel runs cannot corrupt one another's tree.
- Detects the repository's own lint, typecheck, test, and build commands and turns them into gates the agent has to pass. Runs over Codex, Claude CLI, and Pi.

Covenant Compliance Agent Halyk AI Challenge 2026 · document processing

Role: Pipeline design, response-schema validation, release evidence.

- Python pipeline that reads covenant documents, produces structured output, and validates every response against a schema before it counts as an answer.
- Public offline baseline verified at 36/36 — the scoring script and its evidence ship with the repository rather than living in a slide.

AUTOMATION PIPELINES

Artsquare Instagram Pipeline Content archive automation · private source

Role: Pipeline engineering, normalization, test coverage.

- Python pipeline that downloads, normalizes, and locally stores an Instagram archive, with automated tests covering the normalization step.

Tunika publication safeguards Go backend · see page 1

[tunika-tkani.ru](#)

Role: Publication hardening, reconciliation, post-release acceptance.

- Catalog publication with МойСклад integration where incomplete output is rejected instead of replacing the known-good catalog, and Playwright acceptance checks the public site after every release.

AI PRACTICE

Prompt and context pipelines · structured output · tool and function calling · MCP servers · review-before-apply · audit trails · measurable baselines

AI TECHNOLOGY

Python · aiohttp · SQLite · systemd · MCP · Claude Code · Codex · Telegram Bot API · VS Code API · Git worktree